

HILLSBORO, TX, USA

WMO: 720295

Lat: 32.084N

Lon: 97.097W

Elev: 685

StdP: 14.34

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53972

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.7	28.2	11.7	10.3	33.3	15.7	12.5	35.8	28.6	56.9	26.2	53.7	7.1	350	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.1	100.1	72.9	98.6	73.1	96.5	73.5	78.1	88.8	77.3	88.4	76.5	87.6	9.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	137.2	80.7	74.6	132.6	80.7	73.4	127.4	80.7	41.9	88.4	41.1	88.6	40.3	87.6	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.9	22.0	19.4	16.6	103.0	3.5	2.1	14.0	104.5	12.0	105.7	10.0	106.9	7.4	108.4		
(5)			16.3	78.5	3.9	1.5	13.4	79.6	11.1	80.5	8.9	81.4	6.1	82.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.8	47.1	51.1	58.8	65.7	73.1	81.5	84.0	84.9	79.4	68.1	57.7	48.8		
	DBStd	15.51	9.74	10.81	9.31	7.73	6.45	4.07	3.54	4.20	5.60	8.68	9.68	9.63		
	HDD50	500	172	109	33	4	0	0	0	0	0	3	38	141		
	HDD65	2117	557	400	231	83	15	0	0	0	0	69	254	508		
	CDD50	6624	83	140	305	476	716	945	1055	1083	883	565	268	105		
	CDD65	2767	3	11	38	106	266	495	590	618	433	166	34	7		
	CDH74	31027	31	64	217	754	2299	5689	7396	8059	4737	1532	210	39		
	CDH80	15274	1	4	25	174	796	2825	3987	4555	2347	521	36	3		
(14) Wind	WSAvg	9.0	9.0	10.2	11.0	10.9	9.5	8.7	7.8	7.5	7.4	8.5	9.1	8.7		
(15) Precipitation	PrecAvg	39.4	2.9	3.0	3.9	3.4	4.0	3.5	1.9	2.6	3.3	4.3	3.2	3.2		
	PrecMax	54.6	5.4	7.4	10.6	7.3	9.3	10.7	4.6	6.2	10.6	12.6	9.2	9.4		
	PrecMin	23.6	0.5	0.1	0.6	1.2	0.8	0.9	0.0	0.4	0.1	0.6	0.4	0.4		
	PrecStd	8.6	1.4	1.8	2.2	1.9	2.2	2.7	1.3	1.8	2.4	3.1	2.4	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.4	80.6	82.9	88.5	94.6	100.0	101.8	104.0	100.0	93.1	84.7	79.0		
		MCWB	59.6	61.7	64.5	68.8	73.2	73.0	73.2	72.2	70.8	70.9	68.1	63.7		
	2%	DB	72.7	75.6	79.4	84.5	90.3	97.3	99.6	100.4	97.0	89.0	79.4	72.8		
		MCWB	58.5	62.4	65.0	67.5	72.2	73.2	72.7	73.3	72.2	71.5	65.1	62.4		
	5%	DB	68.3	72.0	76.6	81.3	87.7	95.0	97.3	99.0	94.6	85.9	75.6	68.9		
		MCWB	58.4	60.8	64.0	67.2	72.0	73.3	73.3	73.6	72.4	69.2	64.4	60.5		
	10%	DB	63.9	68.3	72.7	78.7	84.5	91.5	95.1	96.9	91.2	82.3	72.5	64.6		
		MCWB	55.9	60.3	62.3	66.4	71.2	73.7	74.0	74.0	71.6	66.8	63.5	57.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.1	69.7	70.0	74.1	77.6	78.5	79.3	78.7	77.6	76.3	72.4	68.5		
		MCDB	70.7	73.5	76.5	81.3	84.7	90.2	87.9	90.4	86.8	83.1	78.5	73.2		
	2%	WB	63.7	66.7	68.4	71.8	75.6	77.3	78.2	77.6	76.6	74.8	69.6	65.8		
		MCDB	68.2	71.2	74.8	80.1	83.6	88.8	88.1	90.4	86.1	82.0	73.9	70.8		
	5%	WB	60.8	64.0	66.7	70.0	74.3	76.4	77.3	76.9	75.5	73.3	67.6	62.7		
		MCDB	66.0	69.3	73.2	77.8	82.6	87.4	88.1	89.8	85.4	81.4	73.3	66.9		
	10%	WB	56.5	61.1	64.7	68.5	72.9	75.5	76.3	76.1	74.6	71.4	65.3	59.3		
		MCDB	63.7	67.0	70.9	76.0	81.5	86.1	87.3	88.9	84.7	79.4	71.3	64.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.1	20.0	20.4	20.5	19.3	19.6	20.3	21.1	20.8	21.8	21.0	20.1		
		MCDBR	27.3	23.9	21.7	21.6	21.7	23.0	23.3	23.7	23.7	22.8	22.8	24.8		
		MCWBR	16.6	14.5	11.7	10.8	7.9	6.0	5.7	5.8	6.9	9.3	12.4	16.0		
		MCDBR	21.7	19.8	18.2	17.7	17.8	18.2	18.9	20.3	18.8	18.7	18.0	20.6		
	5% WB	MCWBR	16.4	14.2	11.9	10.4	8.1	6.0	5.9	5.4	6.2	8.8	13.1	16.5		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.332	0.354	0.396	0.423	0.432	0.439	0.436	0.399	0.361	0.341	0.320		
	taud		2.506	2.459	2.416	2.323	2.304	2.313	2.332	2.320	2.373	2.465	2.467	2.491		
	Ebn at Noon		289	293	294	284	275	272	269	269	275	279	276	280		
	Edh at Noon		27	32	36	41	42	41	40	40	37	31	28	27		
(44) All-Sky Solar Radiation	RadAvg		928	1077	1420	1723	1848	2085	2129	1962	1648	1306	980	817		
	RadStd		93	178	117	141	165	183	148	125	135	193	123	108		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page